André Miranda

Junior Software Developer | Game Programmer (C#, Unity, UE5)

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SUMMARY

Computer Engineering graduate finishing an MSc in Digital Game Development. I've shipped **7 projects** (4 solo, 3 in teams of 2 to 5) in C#, C++ and JavaScript, from gameplay systems and UI to a web app built on a REST API.

TECHNICAL SKILLS

Languages	C#, C++, C, JavaScript, Python, SQL, HTML/CSS, XAML, Matlab
Engines & Tools	Unity, Unreal Engine 5, Git/GitHub, REST APIs
Practices	Object-oriented design, gameplay & UI programming, WebGL/browser builds, responsive web, localisation, user testing

EDUCATION

MSc in Digital Game Development | University of Aveiro — 2024 – present

- Dissertation: humour in microbiology science communication. I designed, programmed and evaluated the game *Superbug* with players, using pre/post-test questionnaires.

BSc in Computer Engineering | University of Trás-os-Montes and Alto Douro (UTAD) — 2019 – 2024

PROJECTS

Superbug | Unity, C# | Solo, MSc dissertation itch.io/superbug — 2026

- Built a top-down action game that serves as the research instrument for my dissertation, and ran an academic study with it.
- Implemented an energy (ATP) economy, a grapple mechanic that steals abilities from enemies, and an ability-assimilation system with time-based decay. Supports keyboard/mouse, Xbox and PlayStation input.

Scoundrel Tower | Unity, C# | Solo itch.io/scoundrel-tower — 2026

- Built the rules engine for a roguelike card game: jokers, shop, weapon forging and bosses that modify the rules.
- Shipped a WebGL build that runs on desktop and mobile browsers, with seeded daily runs, autosave and EN/PT localisation.

MTG Life Counter | JavaScript, HTML/CSS | Solo github.com/Guybota/mtg-life — 2026

- Developed a mobile-first web app for Magic: The Gathering playgroups in about 10,000 lines of framework-free JavaScript: 5 game modes for 2–8 players, deck and player statistics, and an ELO ranking.
- Integrated the Scryfall REST API for card search and art, and Supabase for cloud sync between phones. Installable PWA that works offline through a service worker, covered by 33 Playwright tests.

Unbreakable Phone | Unreal Engine 5 | Team of 5 — 2025

- Programmed the throwing mechanics and the player's special abilities, and designed and implemented all menus and UI for a first-person game.

Newsdle | HTML5 | Team of 2, MSc course project — 2025

- Co-developed a browser game that teaches media literacy across 4 days of news moderation, scored on the balance between audience and credibility.

Volcano Heart | Unity, C# | Team of 4 — 2024 – 2025

- Programmed the boss encounters and the grappling-hook movement for a 2D action-adventure, and did its sound design.

Lords of Conflict | Unity, C# | Solo — 2024

- Programmed a tactics RPG from scratch: grid movement, turn-based combat, unit economy, and configurable map size and player count. Windows and WebGL builds.

CERTIFICATIONS & LANGUAGES

- **Meta** (2023): Programming with JavaScript, Introduction to Front-End Development, Version Control
- **Portuguese** (native), **English** (C2: Cambridge C1 Advanced, grade A, 201/210, 2019)